When Does Aggregating Explanations Work?
PneumoniaMNIST Results

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This report consolidates the released `PneumoniaMNIST` experiment results. Tables are grouped by model, experimental protocol, and aggregation setting. Metric directions and robustness definitions are stated in every table header and caption.

Protocol	Model	Settings
Full-matrix protocol	ViT-B/16	NAIVE, IND, NOISE-S, NOISE-K

Result source snapshot: `d6e4b9bbfb3c` from Workshop-When-Does-Explanations.

Table 1: NAIVE results for `PneumoniaMNIST` with `ViT-B/16` (Full-matrix protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized. The test split contains $n = 624$ samples.

Method	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
	$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Best Individual	0.075	0.301	0.133	0.311	-0.072	0.298	-0.130	0.308	-0.054	0.280	-0.112	0.290	0.030	0.171	-0.024	0.181	0.120	-0.019	0.107	-0.019
Simple Averaging	0.067	0.312	0.090	0.322	-0.064	0.309	-0.087	0.319	-0.051	0.292	-0.064	0.301	0.034	0.183	0.018	0.192	0.085	-0.005	0.079	-0.005
Borda	0.048	0.308	0.077	0.317	-0.045	0.304	-0.074	0.314	-0.032	0.287	-0.058	0.296	0.045	0.178	0.019	0.187	0.106	-0.018	0.099	-0.018
Kemeny–Young	0.042	0.298	0.067	0.308	-0.038	0.295	-0.064	0.304	-0.021	0.277	-0.046	0.287	0.050	0.170	0.024	0.176	0.101	-0.024	0.095	-0.024
RRF	0.050	0.306	0.075	0.316	-0.046	0.303	-0.072	0.312	-0.029	0.285	-0.054	0.295	0.053	0.176	0.027	0.186	0.107	-0.019	0.104	-0.019
Schulze	0.042	0.309	0.074	0.319	-0.038	0.306	-0.071	0.316	-0.022	0.288	-0.054	0.298	0.054	0.179	0.022	0.189	0.107	-0.014	0.095	-0.014

Table 2: IND results for `PneumoniaMNIST` with `ViT-B/16` (Full-matrix protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at $q = 11$. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized. The test split contains $n = 624$ samples.

Method	Setting	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Best Individual	baseline	0.015	0.311	0.031	0.321	-0.017	0.308	-0.028	0.317	-0.019	0.290	-0.015	0.300	-0.011	0.181	0.011	0.191	0.037	-0.011	0.027	-0.011
Simple Averaging	matched NAIVE	0.007	0.316	0.025	0.327	-0.008	0.313	-0.021	0.324	-0.013	0.295	-0.012	0.306	-0.006	0.186	0.010	0.197	0.041	-0.005	0.030	-0.004
Simple Averaging	IND	0.022	0.314	0.035	0.327	-0.024	0.311	-0.030	0.324	-0.022	0.293	-0.026	0.306	-0.005	0.184	0.008	0.197	0.024	-0.005	0.021	-0.002
Borda	matched NAIVE	0.013	0.311	0.025	0.322	-0.014	0.308	-0.021	0.318	-0.022	0.290	-0.012	0.301	-0.012	0.181	0.017	0.192	0.033	-0.012	0.033	-0.011
Borda	IND	0.013	0.319	0.026	0.329	-0.011	0.316	-0.024	0.325	-0.016	0.298	-0.013	0.308	0.006	0.189	0.016	0.199	0.024	-0.005	0.024	-0.005
Kemeny–Young	matched NAIVE	0.014	0.316	0.025	0.325	-0.014	0.312	-0.021	0.322	-0.023	0.295	-0.012	0.304	-0.014	0.186	0.017	0.196	0.034	-0.008	0.035	-0.008
Kemeny–Young	IND	0.013	0.319	0.029	0.329	-0.011	0.316	-0.027	0.325	-0.019	0.298	-0.016	0.308	0.006	0.189	0.019	0.199	0.037	-0.005	0.034	-0.005
RRF	matched NAIVE	0.014	0.312	0.027	0.322	-0.015	0.309	-0.023	0.319	-0.022	0.292	-0.015	0.301	-0.011	0.183	0.016	0.192	0.036	-0.010	0.034	-0.010
RRF	IND	0.014	0.316	0.027	0.325	-0.013	0.312	-0.026	0.322	-0.021	0.295	-0.014	0.304	-0.002	0.186	0.014	0.196	0.027	-0.008	0.024	-0.008
Schulze	matched NAIVE	0.014	0.314	0.027	0.325	-0.014	0.311	-0.023	0.322	-0.022	0.293	-0.013	0.304	-0.012	0.184	0.013	0.195	0.033	-0.009	0.034	-0.007
Schulze	IND	0.019	0.322	0.029	0.332	-0.018	0.319	-0.027	0.329	-0.022	0.301	-0.016	0.311	-0.006	0.192	0.013	0.202	0.022	-0.003	0.022	-0.003

Table 3: NOISE-S results for `PneumoniaMNIST` with `ViT-B/16` (Full-matrix protocol), using the Spearman-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all $\mathtt{R}$ columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Simple Averaging	3	0.080	0.314	0.115	0.324	-0.079	0.311	-0.111	0.321	-0.059	0.293	-0.095	0.303	0.026	0.184	-0.006	0.194	0.079	-0.003	0.062	-0.003
Borda	3	0.077	0.309	0.119	0.319	-0.074	0.306	-0.115	0.316	-0.058	0.288	-0.099	0.298	0.029	0.179	-0.010	0.189	0.123	-0.010	0.117	-0.010
Kemeny–Young	3	0.079	0.304	0.130	0.314	-0.075	0.301	-0.127	0.311	-0.058	0.284	-0.109	0.293	0.024	0.175	-0.024	0.184	0.120	-0.018	0.107	-0.018
RRF	3	0.090	0.304	0.131	0.314	-0.087	0.301	-0.128	0.311	-0.069	0.284	-0.111	0.293	0.014	0.175	-0.024	0.184	0.117	-0.010	0.104	-0.010
Schulze	3	0.082	0.308	0.123	0.317	-0.079	0.304	-0.120	0.314	-0.062	0.287	-0.104	0.296	0.027	0.178	-0.011	0.187	0.122	-0.008	0.115	-0.008

Table 4: NOISE-K results for `PneumoniaMNIST` with `ViT-B/16` (Full-matrix protocol), using the Kendall-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Simple Averaging	3	0.080	0.314	0.115	0.324	-0.079	0.311	-0.111	0.321	-0.059	0.293	-0.095	0.303	0.026	0.184	-0.006	0.194	0.079	-0.003	0.062	-0.003
Borda	3	0.077	0.309	0.119	0.319	-0.074	0.306	-0.115	0.316	-0.058	0.288	-0.099	0.298	0.029	0.179	-0.010	0.189	0.123	-0.010	0.117	-0.010
Kemeny–Young	3	0.079	0.304	0.130	0.314	-0.075	0.301	-0.127	0.311	-0.058	0.284	-0.109	0.293	0.024	0.175	-0.024	0.184	0.120	-0.018	0.107	-0.018
RRF	3	0.090	0.304	0.131	0.314	-0.087	0.301	-0.128	0.311	-0.069	0.284	-0.111	0.293	0.014	0.175	-0.024	0.184	0.117	-0.010	0.104	-0.010
Schulze	3	0.082	0.308	0.123	0.317	-0.079	0.304	-0.120	0.314	-0.062	0.287	-0.104	0.296	0.027	0.178	-0.011	0.187	0.122	-0.008	0.115	-0.008